

Assignment 3 - nanoGPT Model Parallelism on Shakespeare

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Global Settings

- Global batch size: 512
- Sequence length: 256
- Gradient accumulation: 1
- Training Steps: 1000
- Warmup Steps: 20
- Optimizer: AdamW
- Cluster (Delta): Max 6 A100 GPUs

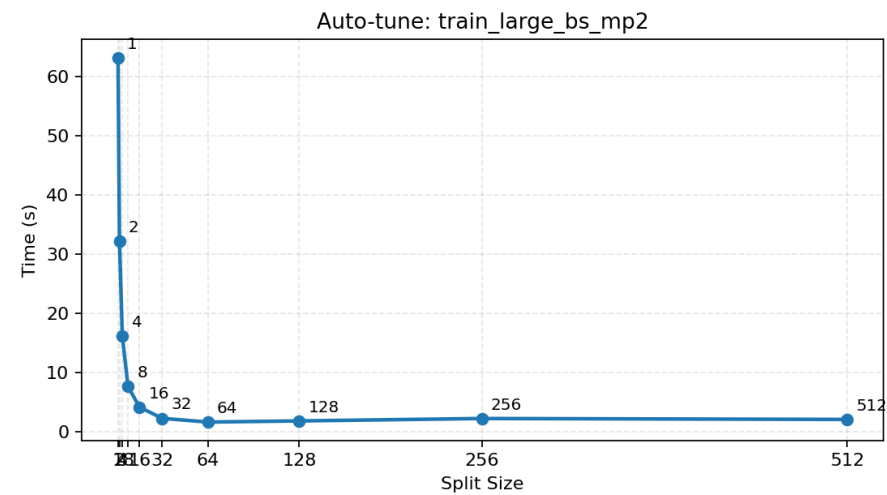
Note that I changed the global batch_size from 64 to 512 to better utilize the GPUs in model parallelism settings. Also, the training steps were reduced from 5000 to 1000 to limit the total runtime for all experiments.

Global Settings (Model Parallel)

- Search Split Size: 1,2,4,8,16,32,64,128,256,512
- Search Warmup Steps: 10
- Search Profile Steps: 20

Task 1: 2-GPU Model Parallelism (MP2)

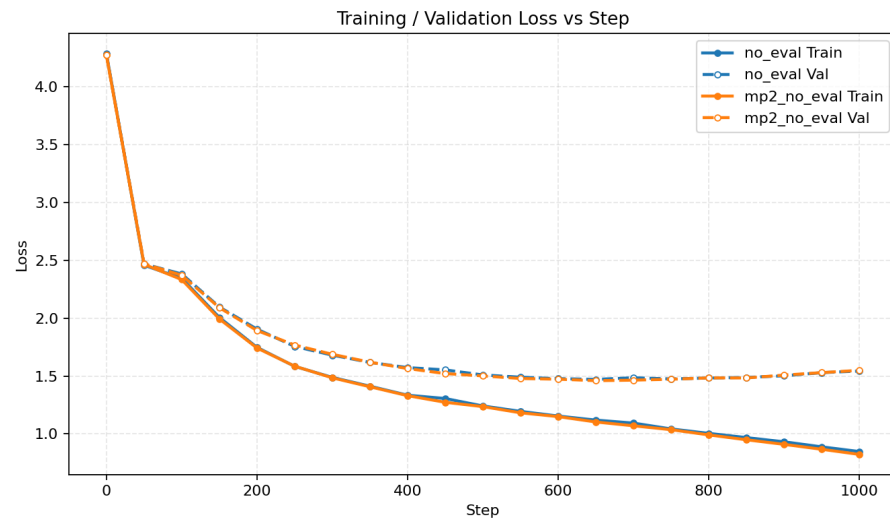
MP2 Split-size Sweep



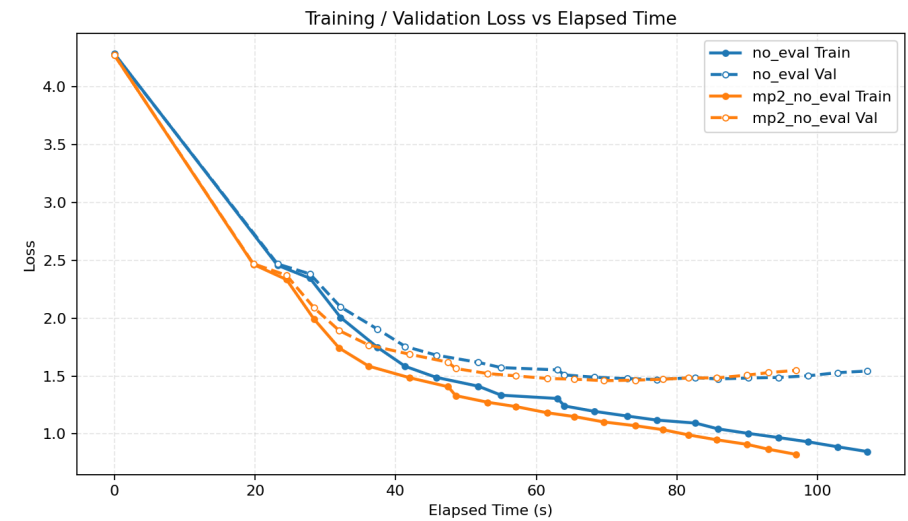
Best split size: 64

The best split size is 64 for MP=2. From the line we can see that very small splits (1,2,4) and very large splits (256,512) both lead to higher step times due to overheads. And the overheads at small splits are more pronounced because of the increased communication frequency.

MP2 vs Baseline: Loss Curves



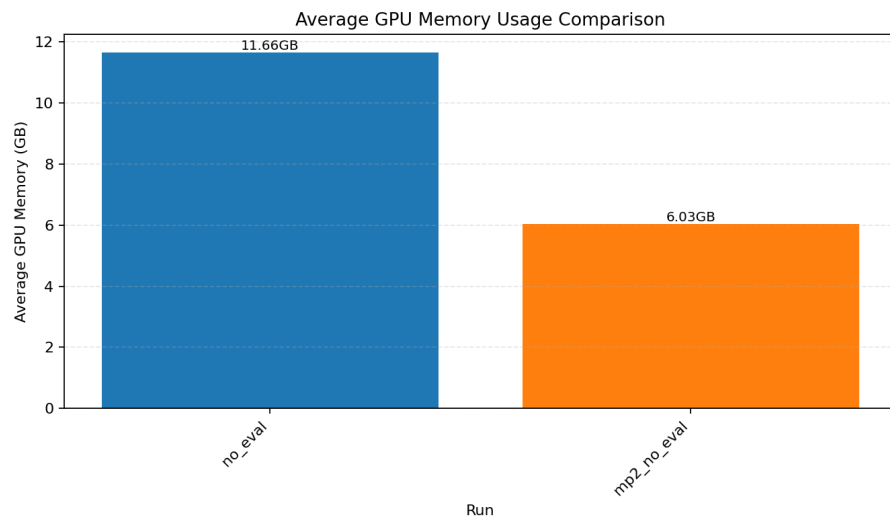
Loss vs steps



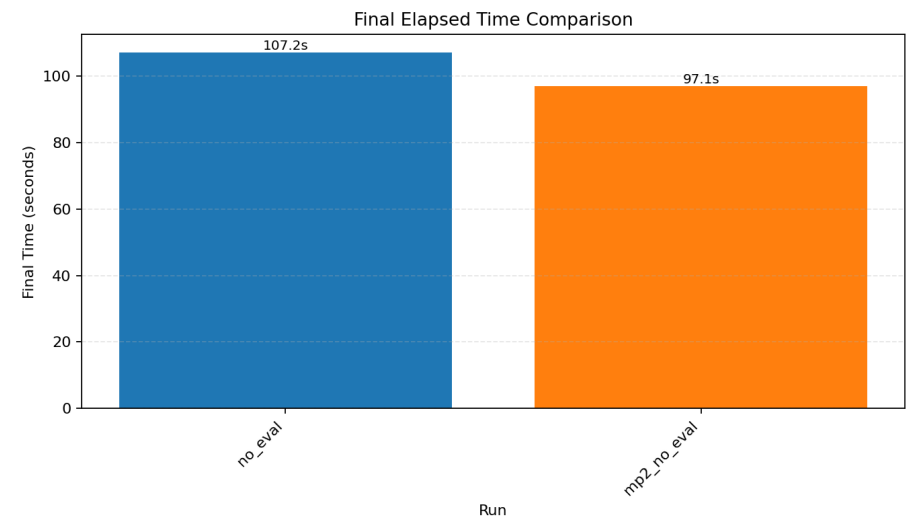
Loss vs time

With the same optimizer and schedule, MP2 reaches the same loss earlier than the single-GPU baseline. When plotted by steps, the two curves closely overlap. This indicates that the optimization is identical.

MP2 vs Baseline: Memory and Runtime



Memory comparison

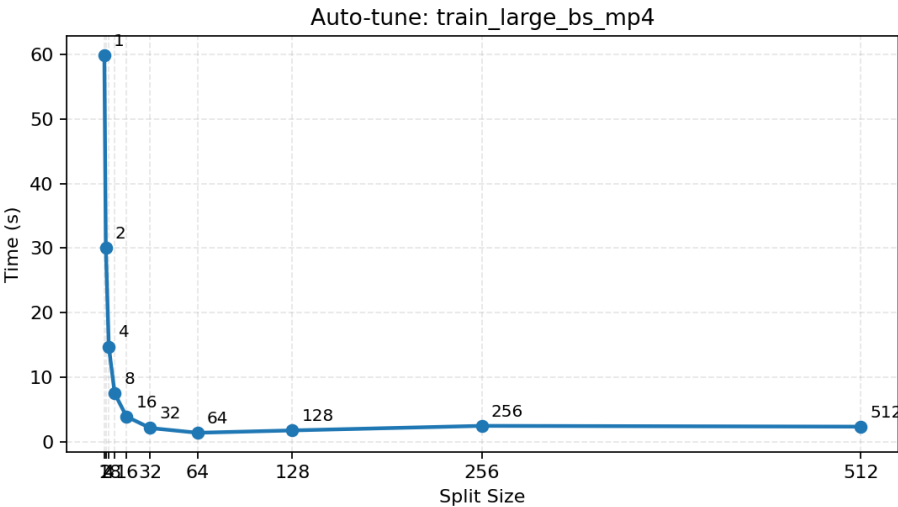


Runtime comparison

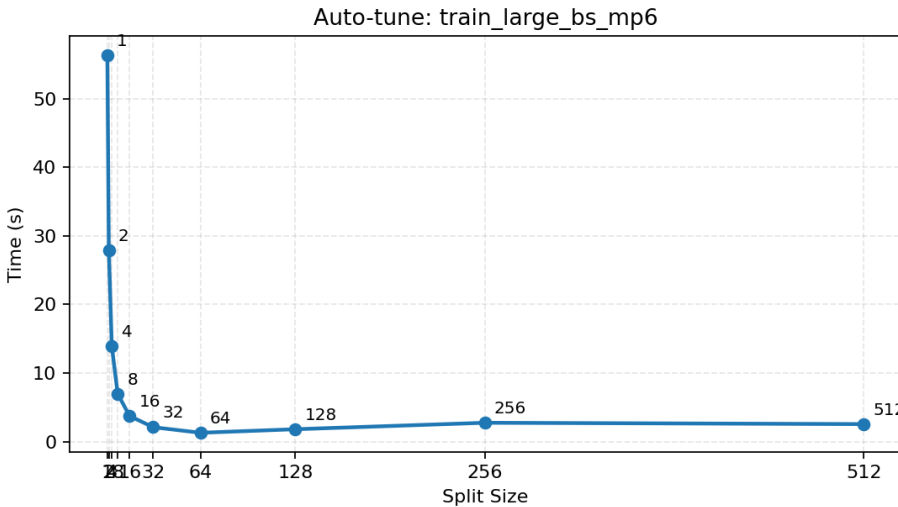
Distributing parameters across 2 GPUs lowers the per-GPU memory peak by half. This leads to a notable reduction in average step time compared to the single-GPU baseline.

Task 2: MP4 & MP6 vs Baseline

MP4 & MP6 Split-size Sweeps



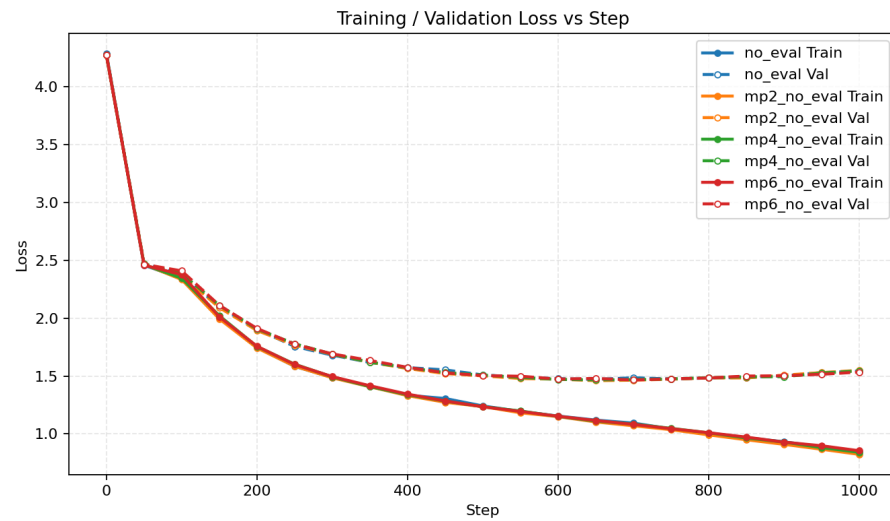
Best split size: 64



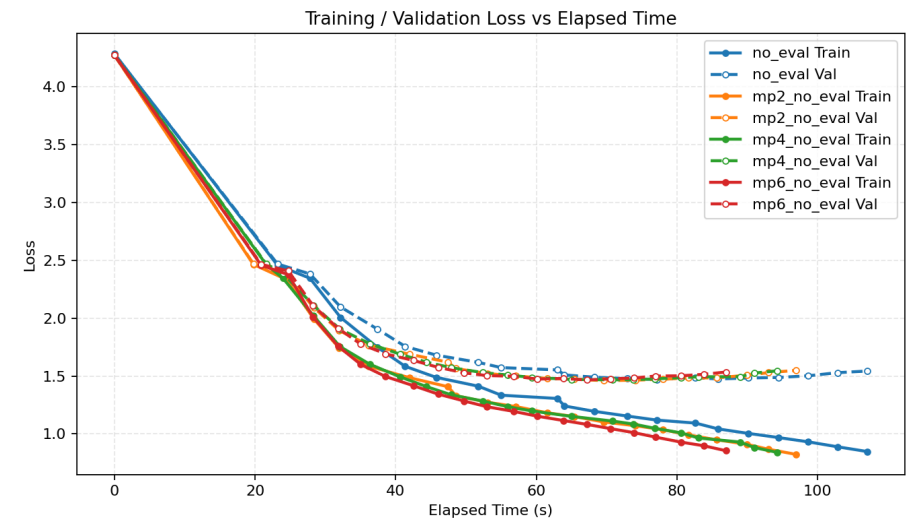
Best split size: 64

The best split size remains 64 when scaling to 4 and 6 stages. This consistency suggests the micro-batch splitting overheads are well-matched at that granularity on this cluster.

Baseline vs MP2/4/6: Loss Curves



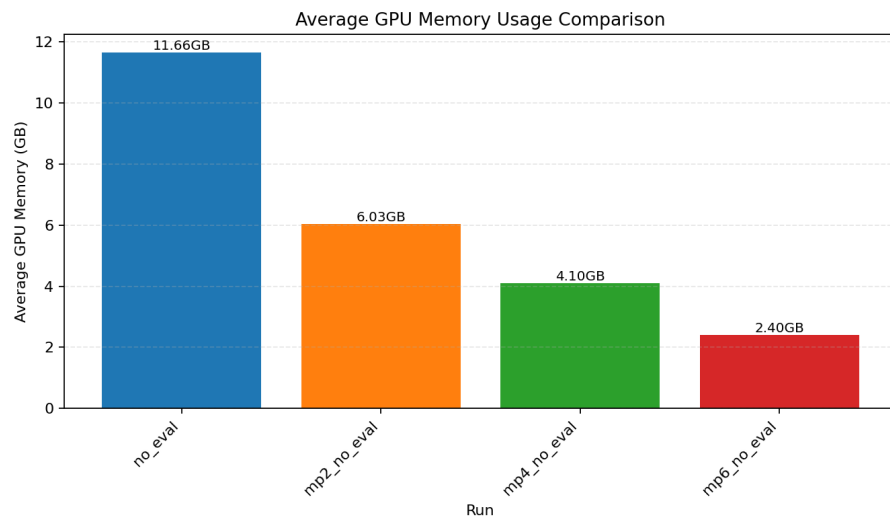
Loss vs steps



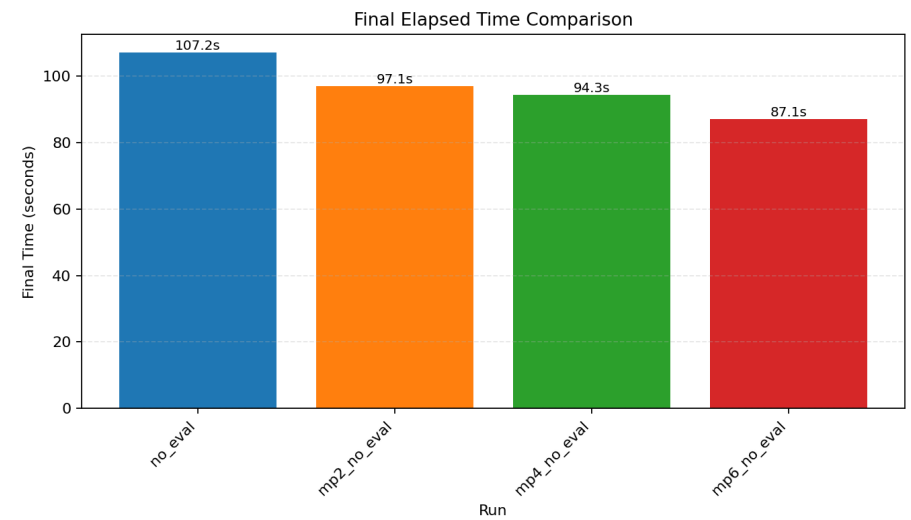
Loss vs time

Moving from 2 to 4 stages pushes the loss-time curve further left, but with sublinear returns relative to stage count. The communication overheads are quite large, as the improvements from 2 to 4 and 6 are quite small.

Baseline vs MP2/4/6: Memory and Runtime



Memory comparison



Runtime comparison

Deeper model parallelism further reduces per-GPU memory pressure and reduces training time.